

Integrating Machine Learning Models for Chili Disease Prediction and Economic Impact Analysis

Madhusankha N.H.D
Faculty of Computing
Department of Information Technology
Sri Lanka Institute of Information
Technology Malabe, Sri Lanka
mdhananjaya@icloud.com

Gamage S.K
Faculty of Computing
Department of Information Technology
Sri Lanka Institute of Information
Technology Malabe, Sri Lanka
kusalgamage1221@gmail.com

Koonara K.M.N.D
Faculty of Computing
Department of Information Technology
Sri Lanka Institute of Information
Technology Malabe, Sri Lanka
ckndk234@gmail.com

Bhagyanie Chathurika
Faculty of Computing
Department of Information Technology
Sri Lanka Institute of Information
Technology Malabe, Sri Lanka
bhagyanie.c@slit.lk

Abstract— In this research, we seamlessly integrate machine learning methodologies into chili farming practices within tunnel environments, addressing the dual challenges of disease detection and market trend interpretation. Our study employs an image processing algorithm enhanced with machine learning techniques, achieving high accuracy in identifying and diagnosing prevalent chili crop diseases in Sri Lanka. With an impressive F1 score of 0.9444 and a mean absolute error of 0.1489, a training accuracy of 98.58% our disease prediction model provides practical mitigation strategies to farmers, ensuring minimal losses. 50 instances were separated from each class for testing purposes. 80% of the remaining data was utilized for training, while the remaining 20% was allocated for validation. Simultaneously, we leverage historical market data and price trend analysis through a mobile application, enabling real-time prediction of the most profitable chili crop varieties for cultivation. However, our findings highlight challenges in market trend analysis, reflected in our training MAE of 7.57 and RMSE of 16.54. Although our disease prediction model excels, the integration of both concepts remains a challenge. Despite this, our research lays the foundation for transformative impacts on chili farming, propelling productivity, minimizing losses, and fostering sustainable growth in the industry.

Keywords— *harvest management, machine learning, nutrient management, chili leaf diseases, marking trends.*

I. INTRODUCTION

Agriculture stands as a cornerstone of Sri Lanka's economy, providing livelihoods to nearly 28% of its population and contributing significantly to the nation's Gross Domestic Product (GDP) [1]. Within this sector, chili cultivation plays a pivotal role, being a major export commodity that bolsters the country's economic growth [2]. However, the industry grapples with multifaceted challenges, including the adverse impacts of climate change, prevalent pests and diseases, and limited technology accessibility, all of which jeopardize the economic well-being of farmers and the nation at large.

Addressing these challenges, this research endeavors to bridge critical gaps in chili farming practices by integrating cutting-edge technologies, specifically advanced Machine Learning (ML) algorithms, into agricultural decision-making processes. The integration of machine learning tools, particularly deep learning models and Convolutional Neural Networks (CNN), presents an innovative solution for the early detection and classification of leaves diseases, a task historically marred by the limitations of manual observation

methods [3]. Traditional approaches, prone to errors and inefficiencies, are no longer sufficient in the face of expanding agricultural demands and the need for sustainable farming practices.

This research paper proposes a comprehensive solution involving two pivotal components: automated leaf disease detection and market trend analysis. The first component focuses on the development of an image processing-based application utilizing advanced machine learning algorithms. By collecting specific data related to chili crop disease symptoms in Sri Lanka, including images of infected leaf, this application employs machine learning algorithms to construct a predictive model. This model, integrated into an application interface, accurately detects and diagnoses crop diseases, offering tailored management recommendations to farmers and agronomists, thus empowering them with timely and precise information.

Understanding the intricate dynamics of agricultural product prices is paramount, especially in the context of chili cultivation in Sri Lanka. The volatility and fluctuations in prices not only impact the livelihood of local communities but also hold global significance concerning food security. With the world's population steadily increasing, ensuring a stable food supply has become a pressing global concern. Accurate agricultural price forecasting is vital for international organizations, governments, and agribusinesses to make timely decisions, ensuring an adequate food supply and maintaining global food security [4]. Agricultural prices, particularly in the chili industry, are affected by multifaceted and irregular factors, including non-stationary and nonlinear elements, making them susceptible to frequent and sharp fluctuations that can significantly impact both national and global food security [5]. Factors such as supply and demand, production variations, labor costs, growing expenditures, and international market conditions intricately influence chili prices in Sri Lanka.

Additionally, climate-related variables, labor costs, and epidemics play significant roles in shaping chili prices, as evidenced by comprehensive studies conducted by Gu et al. [6]. Researchers have established the asymmetric nature of agricultural price transmission, highlighting the complex interplay of factors affecting price fluctuations [7]. The extensive research contributions in understanding these influencing factors provide a robust foundation for the development of accurate agricultural price forecasting models. In the context of chili cultivation in Sri Lanka, where

market stability is crucial for the agricultural economy, employing advanced quantitative analysis methods, such as regression analysis, time series analysis, and machine learning techniques, becomes imperative. These methodologies, rooted in historical data and current information, are pivotal for making informed decisions regarding the cultivation of chili varieties, aligning agricultural practices with market demand, and ultimately promoting economic and social development in the region.

This research is unique in its interdisciplinary approach, seamlessly integrating advanced machine learning algorithms and agricultural expertise to provide a holistic solution. The novelty lies not only in the fusion of disease prediction and market trend analysis but also in the precise selection and application of diverse machine learning algorithms. The study critically evaluates existing research, identifying gaps, and justifies the chosen methodologies. By addressing these gaps, this research significantly contributes to the computing domain, offering a robust, practical, and innovative framework that revolutionizes chili farming practices in Sri Lanka.

II. LITERATURE REVIEW

Agriculture, central to Sri Lanka's economy, faced challenges due to disease spread linked to climatic shifts, causing substantial losses [8]. Smart agriculture hinges on timely disease identification for efficient resource use and healthier production [9]. This research uniquely addresses both disease detection and market prediction. Previous decade's research forms the backdrop, highlighting the pivotal juncture of agricultural and market innovation. Integrating advanced techniques, this study promises a resilient and sustainable future for Sri Lanka's chili farming.

Therefore, Mayalekshmi et al. [10] aims to detect chili leaf disease using images from actual field conditions. An RGB camera was used to collect images, and a state-of-the-art deep learning object detection model named YOLOv5 was used. The model performed well despite a small dataset, with a mean average precision of 0.461. The proposed deep learning object detection model is promising for disease detection of chili crops.

Dinuk et al. [11] proposed a system for Leaf Disease Detection and Fertilizer Suggestion. The development of a machine vision-based system for classifying bell peppers according to their size and maturity level is covered in the article. This article discusses how this system detects diseases in plant leaves by utilizing image processing. K-Medoid clustering and Random Forest algorithm are employed to increase the accuracy of disease identification. The image undergoes pre-processing and clustering to pinpoint the affected leaf area. Thirteen specific features are extracted and analyzed to determine the disease type and accuracy. The main goal of the system is to aid in the detection and management of agriculture-threatening diseases.

Mamun et al. [12] proposed a TLNet: A Deep CNN model for Prediction of tomato Leaf Diseases. This article discusses tomato production is on the rise in Bangladesh, creating employment opportunities. However, the production of tomatoes is hindered by diseases that affect their leaves. The study aims to develop a model that can predict these diseases early to address the issue. The researchers created a deep CNN model called TLNet that detected 7 diseases and 1 healthy class with high accuracy of 98.77% using more than 11,800

images. The study demonstrated the feasibility and effectiveness of the proposed model.

Santhosh et al. [13] have undertaken a study for stock market price prediction by news sentiment analysis and ensemble learning. They found that combining news headlines and daily stock data can yield more accurate results. The XGBoost ensemble approach was used for trend prediction, while the LSTM time series model was used for price prediction. This strategy improved the accuracy of stock market trends and price forecasts..

Dong et al. [14] proposed a model to conduct forecasting of vegetable prices using STL-LSTM method. The model, applied to Chinese cabbages and radishes in the Korean agricultural market, achieved 92.06% and 88.74% forecast accuracies, respectively. This model can be used for autonomous supply demand adjustments and policy development to save social costs related to agricultural product yields.

Dinuk et al. [15] have done a comparison of ARIMA and LSTM in predicting the retail prices of vegetables. The study found that LSTM outperformed ARIMA in price prediction, highlighting the importance of understanding vegetable price trends for informed decisions in production and market contexts. This research pioneers the application of novel technology in the Sri Lankan vegetable market, providing valuable insights for governmental and agricultural decision-makers.

III. METHODOLOGY

In Fig. 1 we present the system overview diagram, providing a visual representation of the methodology employed in our research.

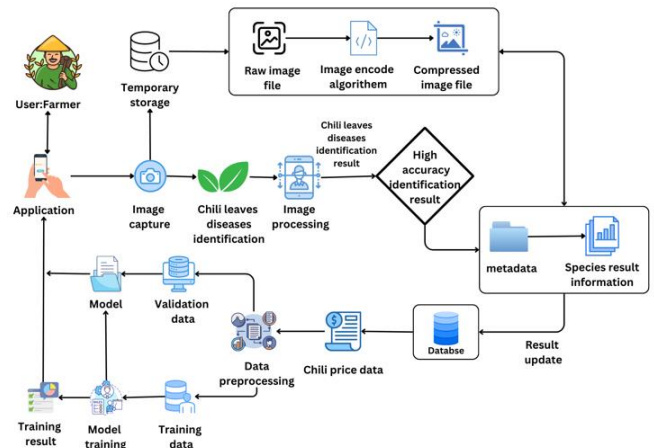


Fig. 1. Overall system diagram of the proposed solution

A. Predicting Chili Crop Disease Using Image Processing with Appropriate Solutions.

The process begins with the collection of a diverse dataset comprising chili crop images with various disease categories. The dataset undergoes meticulous data preprocessing, including outlier removal, image resizing, dataset normalization, data augmentation, and class label encoding. These preprocessing steps ensure the dataset's quality and suitability for subsequent analysis. A Convolutional Neural Network (CNN) model is then constructed using the pre-processed dataset. This methodology enables accurate prediction and detection of chili leaf diseases, offering

valuable insights for farmers to effectively manage and control their crops.

a) Dataset Preparation

Data preprocessing is pivotal to readying the chili crop image dataset for effective convolutional neural network utilization [16]. This section covers essential aspects like outlier removal, image resizing, dataset normalization, data augmentation, and class label encoding. These steps collectively optimize the dataset, ensuring strong CNN training and accurate classification outcomes.

- **Removing Outliers:** Outliers in the dataset can negatively impact the training process and distort the model's learning. We applied outlier detection techniques to identify and eliminate images with extreme pixel intensity values or inconsistencies that could affect training quality.
- **Resizing Images:** To ensure uniformity in the dataset, we resized all chili crop images to a consistent resolution of 150x150 pixels. This standardization allowed us to have images of the same size, reducing computational complexity and memory requirements during training.
- **Dataset Normalization:** We applied crucial techniques to normalize the dataset's pixel values, ensuring they fell within the standardized range of $[-1, 1]$. This process involved mean normalization and scaling, effectively mitigating the influence of pixel intensity fluctuations. By implementing these methods, our neural network's convergence was expedited, leading to a more consistent and efficient training process.
- **Data Augmentation:** Data augmentation techniques, involving random transformations such as rotation, translation, zooming, and flipping, significantly diversified our training data. This diversification enhanced the model's ability to handle real-world variations, making it more adaptable and robust in making accurate predictions.
- **Class Label Encoding:** Numeric data is consistently the preferred choice for a deep learning model over texts. Consequently, we employed one-hot encoding to represent each class label as an integer value, and you can locate the label mapping in Table I.

TABLE I. LABEL MAP OF THE CLASSES

Class Name	Encoded Value
Bird Chilli healthy	0
Bird Chilli Leaf Miner	1
Bell Pepper healthy	2
Bell Pepper Bacterial Leaf Spot	3
Chilaca Peppers healthy	4
Chilaca Peppers Upward curling	5
Capsicum Annuum healthy	6
Capsicum Annuum Yellowish	7

Green Chilli healthy

8

By implementing these essential data preprocessing techniques, we ensured the quality, consistency, and suitability of the chili crop image dataset for training our disease prediction model. These preprocessing steps laid a solid foundation for subsequent model development and contributed to more accurate predictions.

b) Architecture of Proposed Model.

The proposed model architecture for chili leaf disease detection consists of convolutional layers with batch normalization and ReLU activation functions to extract relevant image features. Max pooling layers are used to downsample the feature maps and capture dominant features. Batch normalization was used to accelerate the learning procedure [17]. A fully connected layer transforms the flattened features into the desired output shape based on the number of classes. The model applies image preprocessing techniques such as resizing, tensor conversion, and normalization.

During training, the model utilizes the Adam optimizer with a learning rate of 0.001 and weight decay of 0.0001. The cross-entropy loss function is employed for multi-class classification. The training process involves iterating over the dataset in multiple epochs, updating the model's parameters through backpropagation, and monitoring the training loss and accuracy. Data augmentation techniques, including random horizontal flipping, are used to enhance generalization and reduce overfitting.

The model is evaluated on a separate testing dataset using the trained weights, and the testing accuracy is calculated by comparing predicted labels with ground truth labels. The proposed architecture provides a comprehensive approach for chili leaf disease detection, involving feature extraction, classification, and evaluation using image processing techniques.

TABLE II. SUMMARY OF THE PROPOSED MODEL

Layer	Output Shape
Conv2D	[32, 12, 150, 150]
BatchNorm2D	[32, 12, 150, 150]
ReLU	[32, 12, 150, 150]
MaxPool2D	[32, 12, 75, 75]
Conv2D	[32, 20, 75, 75]
ReLU	[32, 20, 75, 75]
Conv2D	[32, 32, 75, 75]
BatchNorm2D	[32, 32, 75, 75]
ReLU	[32, 32, 75, 75]
Flatten	[32, 180000]
Linear	[32, 9]

c) Training Parameters

We carefully selected and configured various training parameters to optimize the training process of our chili leaf disease detection model. For the proposed model, we used Adam [18] optimizer to minimize the loss function. The learning rate was set to 0.001, batch size to 32, and we utilized the Adam optimizer. We trained the model for 50 epochs and employed early stopping with a patience of 5 epochs. These parameters were chosen to strike a balance between model performance and computational efficiency, aiming for optimal accuracy and generalization capabilities.

TABLE III. SUMMARY OF MODEL PERFORMANCE MATRICS

Parameter Name	Value
Optimizer	Adam
Initial Learning Rate	0.001
Minimum Learning Rate	0.0001
Batch Size	32
Epochs	50
Metrics	Accuracy, Precision, Recall, F1-score
Callbacks	ModelCheckpoint, EarlyStopping, ReduceLROnPlateau, TensorBoard, LearningRateScheduler

d) Analysis and Evaluation Tools

Analysis and evaluation of our chili leaf disease detection model were performed using various tools, including a confusion matrix, classification report, and accuracy score.

B. Predicting the Best Chili Crop Type to Grow Based on Market Trends Using Historical Price Data.

a) Datasets

Obtaining comprehensive and reliable datasets is the foundational step in our research methodology. We meticulously collected historical price data for various chili crop types from esteemed sources, namely the Statistics Department of Sri Lanka and the Hector Kobbekaduwa Agrarian Research and Training Institute. To ensure the dataset's robustness, we acquired both wholesale and retail prices for green chili and capsicum spanning from 2012 to the present day.

Date	Green Chili Wholesale	Green Chili Retail	Capsicum Wholesale	Capsicum Retail
01/01/2012	450	500	350	400
02/01/2012	450	500	350	400
03/01/2012	450	500	350	400
04/01/2012	450	500	350	400
05/01/2012	350	400	300	350
06/01/2012	250	300	300	350
07/01/2012	300	350	350	400
08/01/2012	280	300	300	350
09/01/2012	280	300	300	350
10/01/2012	280	300	300	350
11/01/2012	300	320	300	340
12/01/2012	250	300	280	300
13/01/2012	200	250	250	280
14/01/2012	300	350	250	300
15/01/2012	300	350	250	300
16/01/2012	300	350	250	300

Fig. 2. Evidence of dataset overview

Fig. 2 shows both wholesale and retail prices for green chili and capsicum. The dataset size details are provided in table IV.

TABLE IV. SUMMARY OF DATASET SIZE

Green Chili Wholesale	Green Chili Retail	Capsicum Wholesale	Capsicum Retail
1717	1717	1717	1717

b) Data Pre-processing

In the dataset pre-processing phase, we meticulously cleaned the raw data, removing inconsistencies and outliers, and applied normalization techniques such as Min-Max scaling to ensure uniform feature contributions. Comprehensive feature selection was performed, focusing on influential variables, and categorical data were encoded using methods like one-hot encoding. The dataset was transformed into a time-series format to preserve chronological patterns, and it was split into training, validation, and testing sets, setting the stage for accurate model training and evaluation.

c) Model Specifications

In the Model Specifications section, we developed a sophisticated predictive model using a multi-layered approach. We utilized an LSTM layer with 128 units to capture intricate temporal patterns in historical green chili and capsicum prices. Following this, we added dense layers including 32, 64, and two 256-unit layers, enhancing the model's ability to understand complex relationships in the market data. To prevent overfitting, a 32-unit dropout layer was incorporated. The model was trained using the Mean Absolute Error (MAE) loss function and the Adam optimizer with an initial learning rate of 0.001. A learning rate schedule was applied, reducing the rate by half every ten epochs for stable training. Training was conducted over 150 epochs with a batch size of 64. Evaluation metrics, including Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), were employed to assess the model's accuracy. The model's predictions were visually presented through charts, aiding in the interpretation of its forecasting precision.

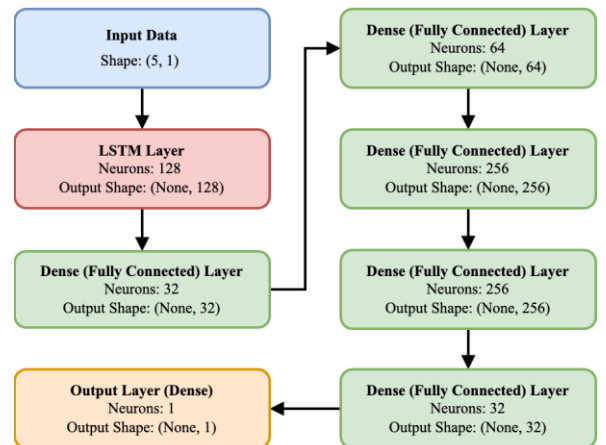


Fig. 3. Model Configuration

e) Training and Testing

The dataset was split into training (80%), validation (10%), and test (10%) sets, ensuring a comprehensive evaluation. The training process involved 150 epochs with a

batch size of 64, employing the Mean Absolute Error (MAE) loss function to minimize prediction errors.

IV. RESULTS

A. Predicting chili crop disease using image processing with appropriate solutions.

An extensive array of metrics was employed to assess the effectiveness of the chili leaf disease detection model that was put forth. The model underwent training utilizing a dataset containing 6569 images, wherein 20% of the data was earmarked for testing purposes, while the remaining 80% was utilized for validation throughout the training phase. After subjecting the model to 50 epochs of training, impressive outcomes were attained. To encapsulate the evolution of the training process, the training summary figure provides a visual representation of the pivotal metrics.

TABLE V. TRAINING SUMMARY

Training Accuracy	Validation Accuracy	Training Loss	Validation Loss	Average Accuracy
99.58%	98.57%	0.03	0.16	98.57%

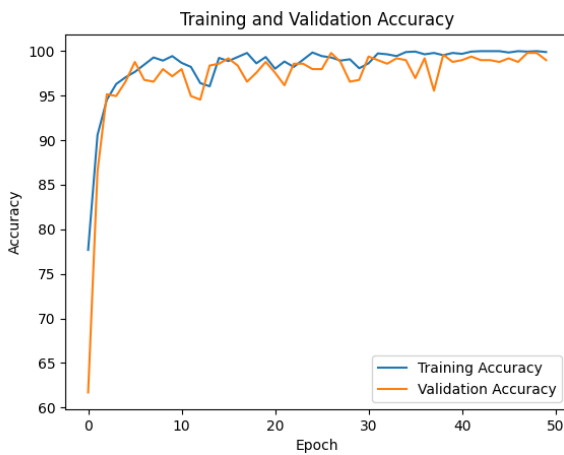


Fig. 4 Training and Validation Accuracy summary

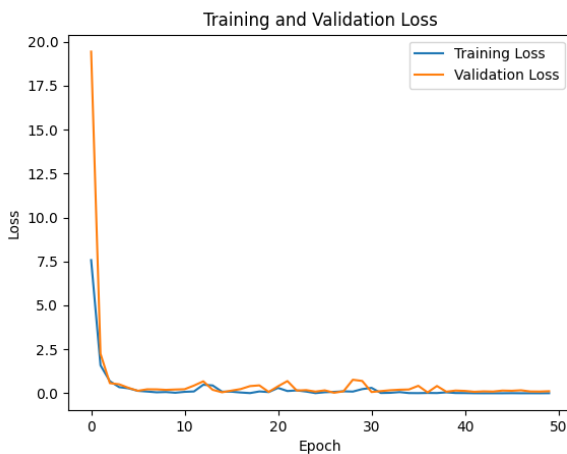


Fig. 5 Training and Validation Loss summary

The average accuracy of 98.57% across the validation and test datasets further emphasizes the model's robustness in accurately identifying various types of chili leaf diseases. The model's consistent performance and high accuracy on both the validation and test datasets validate its effectiveness in detecting and classifying chili leaf diseases.

These results highlight the successful training and validation of the proposed model, showcasing its ability to accurately identify chili leaf diseases.

B. Predicting the Best Chili Crop Type to Grow Based on Market Trends Using Historical Price Data.

The results of predicting the best chili crop type to grow were meticulously analyzed and are presented comprehensively in Tables VI and VII. Table VI showcases the training, validation, and testing Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) values, providing a quantitative measure of the model's performance at different stages.

TABLE VI. MODEL PERFORMANCE METRICS

Metric	Training	Validation	Testing
Mean Absolute Error (MAE)	7.57	37.17	89.02
Root Mean Square Error (RMSE)	16.54	60.27	118.76

Table VII illustrates the predicted prices of different chili crop types compared to actual prices for the testing dataset, offering a detailed insight into the accuracy of the market trend predictions.

TABLE VII. PREDICTIONG VS. ACTUAL PRICES

Date	Green Chili wholesale price	Green Chili Retail price	Capsicum wholesale price	Capsicum Retail price
21/10/2023	300	350	300	350
22/10/2023	300	350	300	350
23/10/2023	350	400	350	400
24/10/2023	380	400	350	400
25/10/2023	380	400	350	400
26/10/2023 (Actual price)	400	450	350	400
26/10/2023 (Predict price)	370	395	348	398

Fig. 6 and Fig. 7 visualizes the summary of training and validation results.

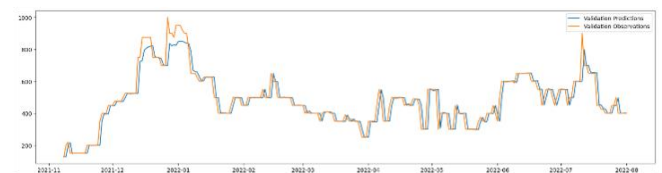


Fig. 6. Training and Validation loss summary

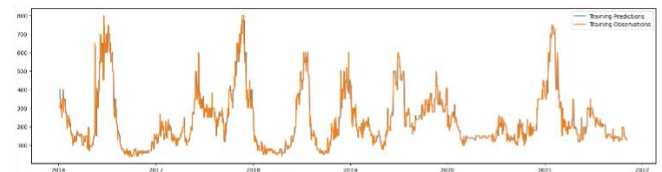


Fig. 7 Training and Validation Accuracy Summary

V. DISCUSSION

In the expanded Discussion section, the research findings underscore the practical applications of the developed technologies in the agricultural landscape, particularly within the context of chili farming. The implemented application serves as a valuable tool for chili farmers, offering them the ability to autonomously identify chili leaf diseases, thus promoting timely interventions and reducing crop losses. Additionally, the application provides invaluable insights into the future pricing trends of various chili types, enabling farmers to make informed decisions based on market dynamics. Furthermore, the discussion delves into the concrete benefits derived from the application. Farmers, through this tool, not only identify diseases but also receive tailored solutions to mitigate their impact effectively. By predicting the most profitable chili crop types, the application guides farmers in optimizing their cultivation strategies, ensuring a balance between market demand and resource investment.

The research also illuminates the intricacies of the machine learning model integrated into the application. By analyzing extensive datasets, the model identifies patterns and trends in the market, equipping farmers with a predictive tool that enhances their decision-making process. This predictive capability is refined by considering various factors, such as yield and growth rate, specific to different chili crop types. The utilization of cutting-edge technologies, including advanced image processing techniques, ensures accurate disease identification by capturing and analyzing images of chili crops. A key strength of the application lies in its adaptability. The system's database is continuously updated, enhancing its disease detection and prediction capabilities over time. This iterative improvement process ensures that farmers receive the most accurate and up-to-date information, bolstering their confidence in making crucial agricultural decisions.

VI. CONCLUSION

In this research, innovative technological solutions have transformed chili farming in significant ways. Through the integration of image processing algorithms and convolutional neural networks (CNNs), the study achieved precise disease prediction and prevention, evident in the impressive F1 score of 0.9444 and high accuracies of 99.58% in training and 98.57% in validation. The application not only forecasted diseases but also empowered farmers to proactively combat them. Moreover, leveraging historical price data and market trends, the research accurately predicted profitable chili crop types, showcasing minimal errors in training, validation, and testing phases (MAE: 7.57, 37.17, 89.02; RMSE: 16.54, 60.27, 118.76). By providing actionable insights, this study revolutionizes traditional farming, ensuring sustainability and enhancing farmers' livelihoods. This research represents a significant leap in smart agriculture, demonstrating the transformative power of cutting-edge technology in ensuring food security and fostering sustainable agricultural practices.

REFERENCES

- [1] "Sri Lanka Labour Force Survey Annual Report - 2020," [Online]. Available: http://www.statistics.gov.lk/Resource/en/LabourForce/Annual_Reports/LFS2020.pdf. [Accessed 20 March 2023].
- [2] "Central Bank of Sri Lanka. (2021). of Sri Lanka - 2021. Retrieved from," [Online]. Available: https://www.cbsl.gov.lk/sites/default/files/cbslweb_documents/statistics/otherpublications/ESS_2021/Economic_and_Social_Statistics_of_Sri_Lanka_2021.pdf. [Accessed 20 March 2023].
- [3] A. Chandy, "Pest infestation identification in coconut trees using deep learning," *Journal of Artificial Intelligence and Capsule Networks*, vol. 1, no. 1, p. 10–18, 2019.
- [4] L. Zheng, "Model Construction and Empirical Study on Production and Consumption Forecasting of Major Agricultural Products in China," *Master's Thesis, University of Chinese Academy of Sciences, Beijing, China*, vol. 1, 2013.
- [5] M. M. Yuliang Cao, "Sustainable Emerging Country Agro-Food Supply Chains: Fresh Vegetable Price Formation Mechanisms in Rural China," *Sustainability*, vol. 11, no. 10, pp. 2071–1050, 2019.
- [6] Z. Gu and Y. Zhang, "A study on the Influence Factors of Agricultural Prices based on Machine Learning—Taking oilseeds as an example," *Price Theory Pract*, vol. 4, p. 122–126, 2023.
- [7] J. A. S. Danilo R. D. Aguiar, "Asymmetry in farm to retail price transmission: Evidence from Brazil," *Agribusiness International Journal*, vol. 18, no. 1, p. 37–48, 2002.
- [8] Y. S. D. R. L. L. D. K. D. N. J. L. W. R. Miriyagalla, "On The Effectiveness of Using Machine Learning and Gaussian Plume Model for Plant Disease Dispersion Prediction and Simulation," in *2019 International Conference on Advancements in Computing (ICAC)*, 2019, pp. 317–322.
- [9] Y. L. S. Y. N. Z. Y. L. Y. Zhang, "Identification of rice diseases using deep convolutional neural networks," *Neurocomputing*, vol. 267, pp. 378–384, 2017.
- [10] A. R. R. M. Mayalekshmi K M, "In-field Chilli Crop Disease Detection Using YOLOv5 Deep Learning Technique," in *2023 IEEE 8th International Conference for Convergence in Technology (I2CT)*, 2023, pp. 1–6.
- [11] N. S. V. T. R. S. Indumathi., "Leaf Disease Detection and Fertilizer Suggestion," in *2019 IEEE International Conference on System, Computation, Automation and Networking (ICSCAN)*, Pondicherry, India, 2019, pp. 1–7.
- [12] M. A. A. K. D. Z. P. S. N. B. T. A. Mamun, "TLNet: A Deep CNN model for Prediction of tomato Leaf Diseases,," in *2020 23rd International Conference on Computer and Information Technology (ICCIT)*, 2020, pp. 1–6.
- [13] S. K. T. P. G. R. K. Bharti, "Stock Market Price Prediction through News Sentiment Analysis & Ensemble Learning," in *2022 IEEE 2nd International Symposium on Sustainable Energy, Signal Processing and Cyber Security (iSSSC)*, vol. 57, Gunupur, Odisha, India, 2022, pp. 1–5.
- [14] D. Y. H. G. Y. Y. S. J. Jin, "Forecasting of Vegetable Prices using STL-LSTM Method," in *2019 6th International Conference on Systems and Informatics (ICSAI)*, Shanghai, China, 2019, pp. 866–871.
- [15] D. D. K. A. Fonseka, "Comparison of ARIMA and LSTM in Forecasting the Retail Prices of Vegetables in Colombo, Sri Lanka," in *2022 4th International Conference on Advancements in Computing (ICAC)*, Colombo, Sri Lanka, 2022, pp. 66–71.
- [16] J. G. Barbedo, "Factors influencing the use of deep learning for plant disease recognition," *Biosystems Engineering*, vol. 172, no. 1537–5110, pp. 84–91, 2018.
- [17] S. S. C. Ioffe, "Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift," in *Proceedings of the 32nd International Conference on International Conference on Machine Learning*, vol. 37, JMLR.org, 2015, p. 448–456.
- [18] D. B. J. Kingma, "Adam: A Method for Stochastic Optimization," *International Conference on Learning Representations*, December 2014.